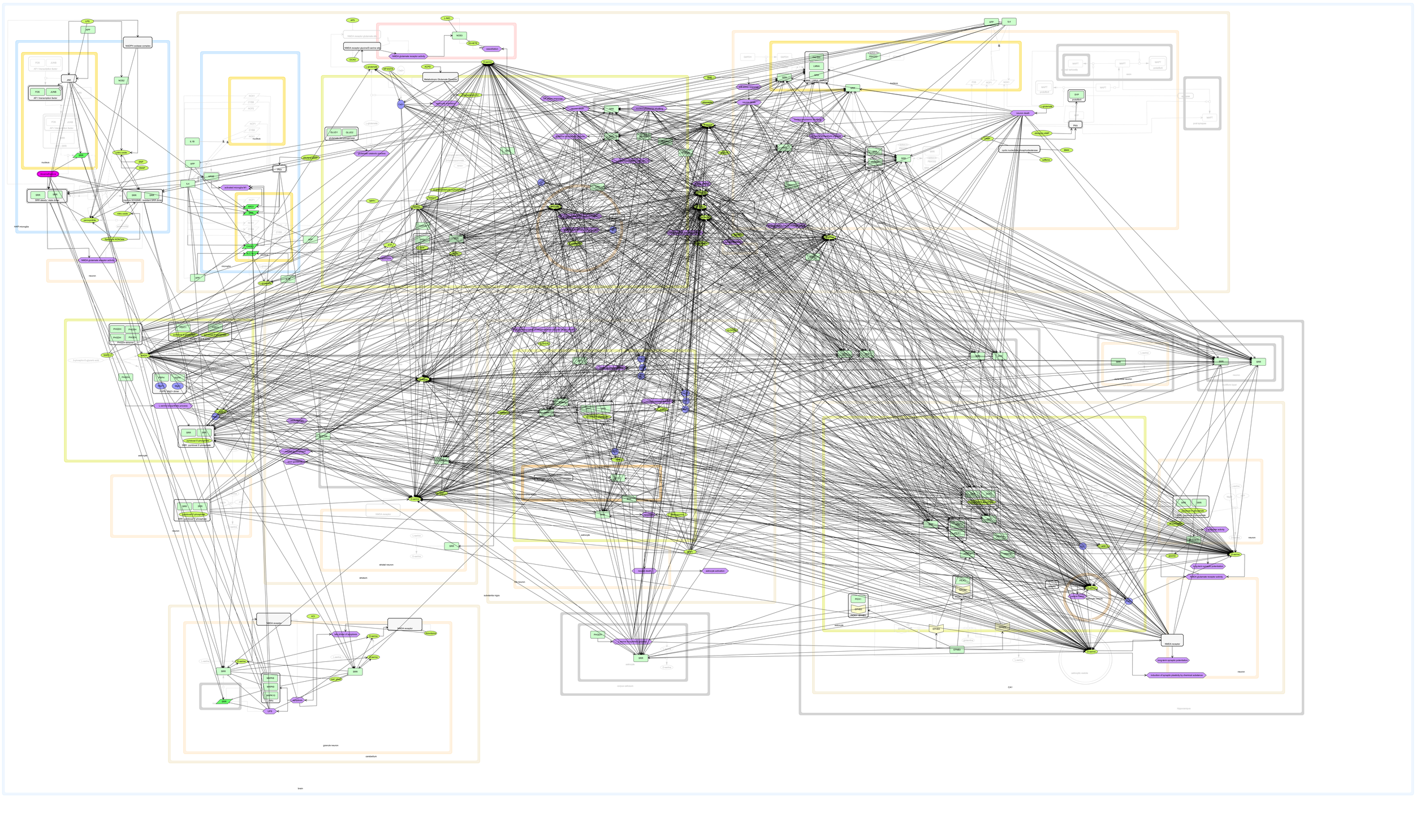

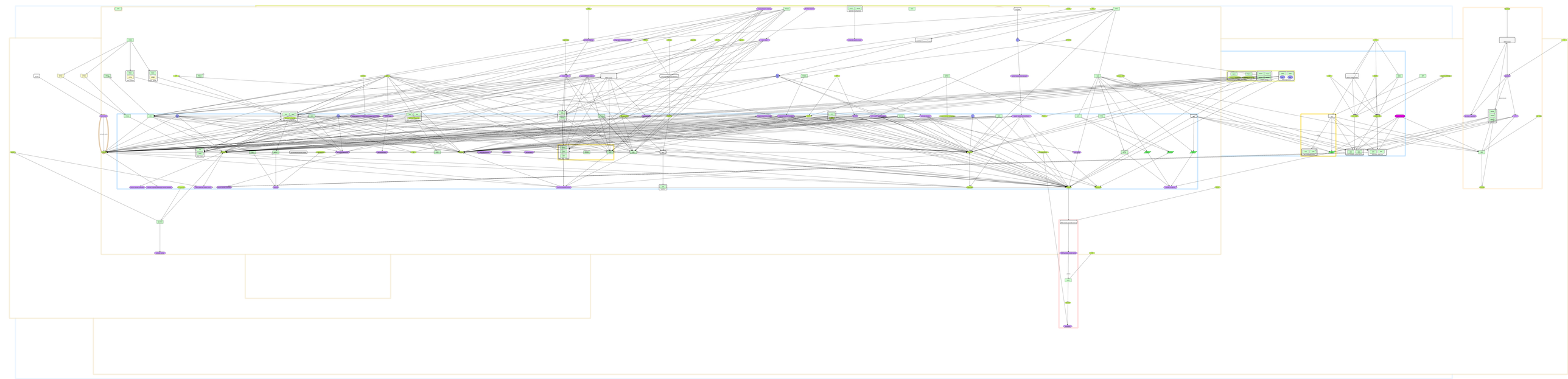

LPS
APP
NADPH oxidase complex
FOS
JUNB
AP-1 transcription factor
JNK
NOS2
P
FOS
JUNB
AP-1 transcription factor
SRR
AP-1 : SRR
SRR
nucleus
nitric oxide
SNP
SNAP
dexamethasone
SRR
SRR
SRR steady state dimer
SRR
SRR
inactive SDS/βME resistant SRR dimer
peroxynitrite
nitric oxide
superoxide
Spermine NONOate
HAPI microglia
NMDA glutamate receptor activity
neuron
AP5
NMDA receptor glutamate site
NMDA receptor glycine/D-serine site
DCKA
NMDA glutamate receptor activity
L-NIO
NOS3
20-HETE
arte
vasodilation
D-serine
L-glutamate
NP-EGTA
Ca2+
ACPD
Metabotropic Glutamate Receptor
NOX1
CYBB
NCF2
NCF1
CYBB
NOX1
nucleus
IL1B
&
&
GLUD1
GLUD2
glutamate dehydrogenase
glutamate catabolic process
APP
dibutyryl cAMP
PKA
APOE
IL4
activated microglia M1
NCF1
ARG1
SRR
CHI3L1
SLC7A11
nucleus
microglia
APP
L-glutamate
IL1B
APP
prostaglandin E2
astrocyte activation
hydrogen peroxide
3-hydroxypyruvate
NH4+
D-glyceraldehyde 3-phosphate
NAD(+)
MPP+
NADH
GAPDH
SRR
GAPDH : SRR
SRR
ATP
SRR : ATP
ATP
glycolysis
ER stress response
neuron death
SRR
nucleocytoplasmic shuttling
D-serine biosynthetic process
SRR
FBXO22
PRKCA
DAO
L-serine biosynthetic process
BME
etoposide
ER stress response
neuron death
D-serine
GAPDH
GAPDH
LMNA
GAPDH
LMNA
SRR
SRR
SRR : LMNA : GAPDH
isoform 1
FBXO22
nucleus
SRR
nucleocytoplasmic shuttling
D-serine biosynthetic process
SRR
SRR
isoform 1
FBXO22
SRR : FBXO22a
SRR
isoform 1
FBXO22
isoform 2
FBXO22
SRR : FBXO22b
Cl-
SRR
D-serine
D-serine
D-serine
NMDA receptor
SNARE
PRKCA
neuron
APP
IL4
&
FOS
NCF1
NOX1
MAPT
axonal varicosity
MAPT
MAPT
protofibril
axon
MAPT
protofibril
MAPT
SYP
protofibril
synapse
SYP
PKA
MAPT
post synapse
L-glutamate
neuron death
dibutyryl cAMP
cAMP
cyclic nucleotide phosphodiesterase
IBMX
caffeine
SRR
FBXO22
SRR : FBXO22b
PHGDH
PHGDH
PHGDH
PHGDH
PHGDH tetramer
PSAT1
PSAT1
pyridoxal 5'-phosphate
pyridoxal 5'-phosphate
PSAT1 : PXLP dimer
NAD(+)
NADH
3-phospho-D-glyceric acid
L-glutamate
PHGDH
PSPH
PSPH
Mg2+
Mg2+
PSPH : Mg2+ dimer
L-serine biosynthetic process
ATP
Mg2+
microcephaly
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
SLC1A4
neuron development
axon guidance
astrocyte
L-serine
L-serine
D-serine
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
Mg2+
ATP
D-serine
neuron
glutamatergic neuron
D-serine
NMDA receptor
L-serine
SRR
D-serine
striatal neuron
striatum
positive regulation of oxidative stress-induced neuron death
6-OHDA
6-OHDA
SRR
serine racemase activity
Mg2+
Mg2+
Mg2+
MPP+
SRR
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
serine racemase activity
ATP
Mg2+
Mg2+
Mg2+
Mg2+
ATP
pyruvate dehydrogenase complex
BCL2
BAX
mitochondrial matrix
SRR
apoptosis
staurosporine
astrocyte
SRR
MPP+
neuron death
astrocyte activation
DA neuron
substantia nigra
PHGDH
L-serine biosynthetic process
L-serine
SRR
astrocyte
D-serine
corpus callosum
SRR
SRR
SRR
SRR
L-serine
D-serine
pyramidal neuron
CA fields
neuron
multiform layer
subiculum
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
SRR
SRR
SRR
PICK1
SRR : PICK1
PRKCA
PRKCA
PRKCA
PICK1
EPHB3
PICK1 : EPHB3
PICK1
EPHA4
PICK1 : EPHA4
EPHB3
EPHB3
EPHA4
glutamine
EFNB3
L-serine
astrocyte
Mg2+
ATP
SNARE
D-serine
vesicle fusion
astrocytic vesicle
Ca2+
D-serine
astrocytic vesicle
L-serine
Mg2+
ATP
H2O
NH3
pyruvic acid
SRR
SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
D-serine
neuron
D-isoleucine
transporter activity
SLC7A10
glycine
D-serine
long-term synaptic potentiation
NMDA glutamate receptor activity
NMDA receptor
long-term synaptic potentiation
induction of synaptic plasticity by chemical substance
neuron
CA1
hippocampus
NMDA receptor
KCl
NMDA receptor
dizocilpine
late phase of apoptosis
D-serine
L-serine
D-serine
L-serine
D-serine
SRR
SRR
MAPK8
MAPK9
MAPK10
JNKs
met‥phen
SRR
nucleus
apoptosis
UPS
granule neuron
cerebellum
brain

SRR
EFNB3
SNARE
EPHA4
EPHB3
PRKCA
PICK1
EPHA4
PICK1 : EPHA4
PICK1
EPHB3
PICK1 : EPHB3
ATP
PRKCA
6-OHDA
MPP+
BME
nucleocytoplasmic shuttling
FBXO22
ER stress response
GLUD1
GLUD2
glutamate dehydrogenase
DAO
V-ATPase
ACPD
APS
TGFB1
dizocilpine
L-glutamate
ER stress response
voltage-gated sodium channel activity
IBMX
caffeine
etoposide
MK-801
glycine
neuron death
glutamate catabolic process
Metabotropic Glutamate Receptor
H+
NP-EGTA
LPS
NMDA receptor
KCl
neuron death
nucleocytoplasmic shuttling
NMDA receptor
cyclic nucleotide phosphodiesterase
NO
PHGDH
MPP+
GAPDH
proton transmembrane transport
IL4
dibutyryl cAMP
PSAT1
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pyridoxal 5'-phosphate
pyridoxal 5'-phosphate
PSAT1 : PXLP dimer
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PHGDH
PHGDH
PHGDH tetramer
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Mg2+
PSPH : Mg2+ dimer
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Spermine NONOate
apoptosis
granule neuron
vesicle fusion
astrocyte vesicle
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Mg2+
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SRR
pyridoxal 5'-phosphate
SRR : pyridoxal 5'-phosphate
SDS
Mn2+
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SRR : pyridoxal 5'-phosphate
SRR
isoform 1
FBXO22
SRR : FBXO22a
PRKCA
S657
P
dibutyryl cAMP
synaptogenesis
cAMP
D-serine biosynthetic process
L-serine biosynthetic process
NADH
SRR
ATP
SRR : ATP
glycolysis
TGFB receptor signaling pathway
SLC1A4
astrocyte activation
D-glyceraldehyde 3-phosphate
O2-
SRR
synaptic vesicle lumen acidification
NAD(+)
APP
IL1B
PKA
JNK
nucleus
nitric oxide
peroxynitrite
dexamethasone
late phase of apoptosis
MAPK8
MAPK9
MAPK10
JNKs
UPS
met□phen
glycine
D-serine
SRR
PICK1
SRR : PICK1
D-serine
BAX
BCL2
S70
P
pyruvate dehydrogenase complex
staurosporine
serine racemase activity
astrocyte activation
13J
SRR
D-serine
neuron development
microcephaly
axon guidance
GAPDH
LMNA
SRR
SRR : LMNA : GAPDH
FBXO22
SRR
SRR
microglia
PKA
L-serine
prostaglandin E2
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L-glutamate
activated microglia M1
DCKA
D-serine
SLC7A10
antiporter activity
NMDA receptor glycine/D-serine site
NMDA glutamate receptor activity
L-NIO
arteriole
NOS3
20-HETE
vasodilation